

Safety AARL: Weight adjustment for reinforcement-learning-based safety dynamic asset allocation strategies

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ARTICLE INFO

Keywords:

Reinforcement learning
Dynamic asset allocation
Proximal policy optimization
Weight adjustment
Stock trading

ABSTRACT

Dynamic asset allocation involves adjusting asset weights based on performance to reduce risks according to market conditions. This paper proposes a novel framework called safety asset allocation reinforcement learning (AARL), which combines multiple protective dynamic asset allocation strategies (PDAS) to minimize risks through reinforcement learning (RL). The framework integrates eight validated PDAS that maintain their unique characteristics. To implement the proposed framework, a novel RL environment and agent were developed to invest in PDAS using three high-probability actions. The agent receives state and reward information from the environment and is trained using the proximal policy optimization algorithm to output weights for investing. The actions are PDAS, which facilitates for diversified investments and the learning of various combinations that are suitable for market conditions. Three experiments were conducted to verify the effectiveness of the proposed Safety AARL framework. The first experiment demonstrates the validity of adjusting weights using the framework. The second experiment demonstrates that the framework outperforms existing techniques during global market decline. The third experiment demonstrates that risk is minimized, outperforming the latest methods for asset allocation. Overall, the results demonstrate that the Safety AARL framework outperforms existing techniques by minimizing risks and maintaining stable returns by combining the advantages of multiple PDAS.

1. Introduction

A portfolio is a collection of financial assets including stocks, bonds, currencies, and commodities such as gold. Asset allocation is a crucial area of research in finance that involves assigning different weights to assets in a portfolio to minimize risk. Diversification across assets with low correlations is an effective method of preventing losses, even when the returns on other assets decline (Markowitz, 1952). Diversification can also be applied to investment strategies aimed at maximizing portfolio profits while minimizing risk (Almahdi & Yang, 2017). Given that most individuals are more sensitive to risk than investment returns, the goal of minimizing risk is paramount (Tversky & Kahneman, 1992). Personal decision-making plays a critical role in portfolio investments in the finance market (Yang, 2019; Cheong et al., 2017).

Stock trading and asset allocation trading are sophisticated financial activities that necessitate expertise, experience, and analytical abilities to make optimal investment decisions. Traders have traditionally employed statistical models and technical analysis to predict market

trends and maximize returns. However, these methods have limitations, particularly considering the increasing volume and complexity of financial data. Reinforcement learning (RL) is a machine learning technique that has the potential to enhance investment decision making in stock trading and asset allocation trading by providing a data-driven, adaptive approach to investment strategies. In recent years, there has been growing interest in applying artificial intelligence (AI) techniques in the financial sector, including big data analysis and machine learning algorithms, in various fields such as asset price prediction, risk analysis, and program transactions (Hu & Lin, 2019; Chen et al., 2021; Ponomarev et al., 2019; Pendharkar & Cusatis, 2018; Lei et al., 2020; Park et al., 2020; Yang et al., 2018; Lee, 2019; Zhang et al., 2020; Hirata et al., 2016; Prollochs et al., 2016). In particular, research on asset allocation has focused on direct stock transactions and asset allocation transactions based on RL over the past two decades (Heaton et al., 2017).

The majority of RL-based asset allocation studies utilize an architecture for adjusting the investment weights of assets and can be broadly categorized into two approaches: RL internal function conversion and

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the fusion of other techniques. Although both methods have outperformed previous tools (Lee et al., 2020; Li, Zhang, & Chan, 2007; Jiang et al., 2017; Hu & Lin, 2019; Soleymani & Paquet, 2020; Ye et al., 2020; Betancourt & Chen, 2021), they still face significant issues. First, as the number of assets considered for investment increases, the number of RL actions also increases proportionally, making optimization challenging and limiting the potential for diversified investments. Second, most studies have been limited to input variables and reward functions, despite a large number of attempts to utilize previously developed asset allocation strategies. Finally, although some studies have achieved higher returns, they have also resulted in higher risk.

To address these concerns, we propose a novel framework called safety asset allocation reinforcement learning (AARL) that minimizes risks by combining protective dynamic asset allocation strategies (PDAS) based on RL. Accordingly, to extract variables using an autoencoder and maintain the characteristics of PDAS, we define strategies as RL actions and create separate dynamic asset allocation functions corresponding to actions in the RL environment for interaction. The main contributions of this study can be summarized as follows.

- 1) The Safety AARL framework introduces a new approach to AARL by providing an RL environment and agents to enable the use of multiple PDASs in combination depending on market conditions. This is a different approach to previous studies that use RL to create a single PDAS.
- 2) Previous studies have been limited to using PDAS for input variables and output layers. However, the proposed framework extends the use of PDAS by introducing a method to calculate rewards by matching each action with a PDAS built in the RL environment. This is a novel approach to fully exploit PDAS.
- 3) In addition, previous studies have suffered from the problem that the number of RL actions grows proportionally with the number of assets to be invested. The proposed framework compensates for this by defining a PDAS as an action and increasing the number of assets belonging to the action.
- 4) Three experiments demonstrate that the returns achieved by the proposed model are stable on an upward slope and that risk is minimized, regardless of market conditions, by combining the advantages of different PDAS. Compared to the most recently developed existing models, the proposed framework results in a lower rate of return, but a 2.506 and 20.594 times higher Sharpe ratio and maximum drawdown (MDD), respectively, leading to superior overall performance.

The remainder of this paper is organized as follows. Section 2 presents related background knowledge and discusses related works. Section 3 described the proposed safety AARL framework. Section 4 presents our experimental results. Finally, Section 5 concludes this paper.

2. Background

2.1. Dynamic asset allocation

DAS attempt to select suitable asset classes depending on market conditions or adjusting the ratios of assets. Weight adjustment involves reducing the weight of assets with poor performance while increasing the weight of assets with favorable performance. The general premise of DAS is to prepare for risk and falls, and to achieve optimal returns based on market trends. The advantages of DAS include high return rates, low MDD, and portfolio diversity. Drawbacks include the possibility of over-optimization, charges from frequent transactions, and complexity in portfolio management. Well-known PDAS include dual momentum (DM), composite DM (CDM), global tactical asset allocation 5 (GTAA5), global tactical asset allocation aggressive 3 (GTAA Aggressive 3), protective asset allocation (PAA), vigilant asset allocation G4 aggressive

(VAA-G4 Aggressive), vigilant asset allocation G12 (VAA-G12), and defensive asset allocation (DAA).

1. DM (HA & Fabozzi, 2022) involves investing in stocks in the United States or other advanced countries based on 12 month relative momentum. Stocks are sold and bonds are purchased if the earnings over the past 12 months in two stock markets are below the corresponding cash values.
2. CDM (HA & Fabozzi, 2022) focuses on assets that are fundamentally robust, stocks, bonds, real estate, and recessions, where a portfolio is divided into four parts and then the MDD is lowered once. Then, the absolute momentum technique is used to lower the MDD again to avoid investing in a bear market.
3. GTAA5 (Faber, 2007) applies absolute momentum, where a portfolio is divided into five asset classes and assets are purchased if the price of each asset is above the 12 month moving average. Otherwise, cash is held.
4. GTAA Aggressive 3 (Faber, 2007) applies absolute momentum and relative momentum to asset allocation. The average return of each exchanged traded fund (ETF) is calculated and then the three ETFs with the highest returns are purchased at the same price. Cash is held if the price of each ETF is lower than the 10 month moving average.
5. PAA (Keller & Keuning, 2016) involves buying ETFs with a recent price increase, which are chosen from 12 ETFs based on relative momentum, and the weight for investing in risk-free assets is determined based on breadth momentum.
6. VAA-G4 (Keller & Keuning, 2017) involves buying ETFs with a recent price increase, which are chosen from four ETFs based on relative momentum. The weight of a recent profit increases significantly with respect to relative momentum and breadth momentum is used similarly to PAA.
7. VAA-G12 (Keller & Keuning, 2017) is similar to VAA-G4, but invests in five assets among the 12 asset classes for the purpose of diversified investment.

DAA (Keller & Keuning, 2018) is similar to VAA, but proposes developing domestic and US mixed bonds as a canary universe. Risks are avoided by calculating the weights of assets in the canary universe.

2.2. Reinforcement learning (RL)

RL is a machine learning technique used in various application programs and is inspired by behavioristic psychology. RL consists of an agent, environment, state, action, and reward. The agent perceives the current state in an environment and selects the action that maximizes the reward from among multiple possible actions. This process is illustrated in Fig. 1. Such a learning method gradually improves the overall performance a model based on experience gained through trial and error. Essentially, RL is based on a Markov decision process (MDP) in which the current state is considered to be related to the past (Van Otterlo & Wiering, 2012). This MDP property can also be observed in financial markets. The efficient market hypothesis states that past information is efficiently reflected in current prices (Malkiel, 1989). Therefore, future information can be extracted from current information, implying that RL and the MDP are appropriate and share the same perspective as a finance market (Wu et al., 2021).

The proximal policy optimization (PPO) algorithm, which is a revised version of the trust region policy optimization (TRPO) algorithm, is widely used in RL applications (Schulman et al., 2017; Schulman et al., 2015). The PPO algorithm uses two types of network called as actor and critic that are updated optimally. The actor network proposes the best policy for the agent based on the current state input and the critic network evaluates policy quality and suggests potential improvements. PPO has been shown to provide better sample efficiency and performance compared to other RL algorithms. Therefore, PPO is a widely employed in diverse fields, including robotics, gaming, and

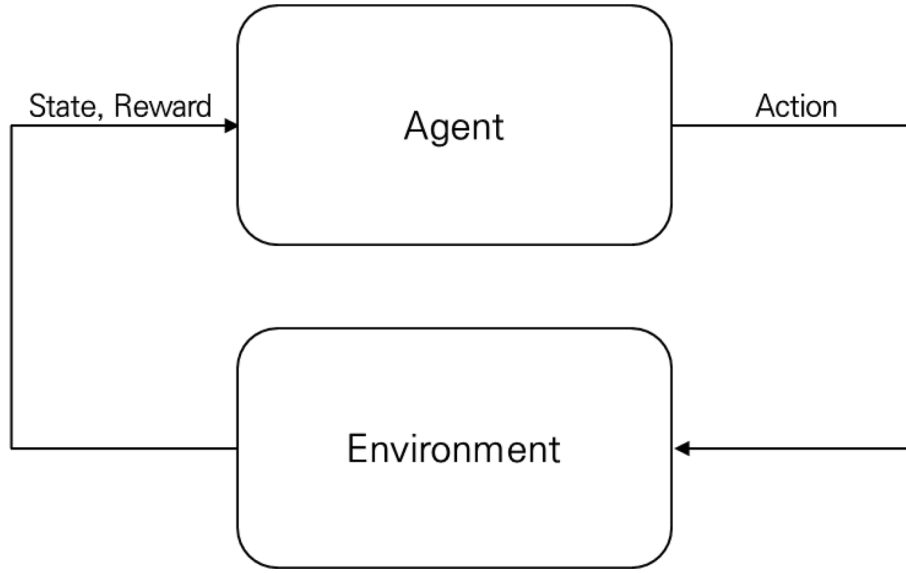


Fig. 1. Basic architecture of reinforcement learning (RL): an agent, environment, state, action, and reward.

finance. The objective of general RL is to find a parameter (θ) representing the policy that can maximize the expected reward. The objective function, which defines the expected reward, can be expressed as

$$\hat{g} = \hat{E}_t[\sum \log \pi_\theta(a_t|s_t) \hat{A}_t] \quad (1)$$

where π_θ is the stochastic policy and $\hat{A}_t$ is the advantage estimated at time t by the network evaluating values. To maximize the objective function, a parameter (θ) is optimized using a gradient in RL. Conventional TRPO incurs a high computational cost because it uses a conjugate gradient or second-order KL-divergence when solving a constraint optimization problem. In contrast, PPO reduces the computational cost by setting a confidence interval using the clipping technique defined in Eq. (2) when updating the surrogate function. ϵ , which is a hyperparameter, yields the best performance with a value of 0.2 for continuous actions and a value of $0.1 \times \alpha$ for discrete actions. The learning rate α starts at one and converges to zero. When the advantage is positive, the next value is higher than the current value and this parameter is updated in the positive direction. The probability $\pi(a_t|s_t)$ of selecting action a in state s is updated to increase. If $r_t(\theta)$ increases infinitely, it is clipped by ϵ to update the parameter within the confidence interval. When the advantage is negative, the next value is lower than the current value and the parameter is updated via clipping within the confidence interval to reduce the probability $\pi(a_t|s_t)$ of selecting action a in state s . PPO is an advantage actor critic algorithm that is updated in the same manner as an advantage actor. PPO updates the actor network to maximize the surrogate function (L^{CLIP}) and action entropy ($S[\pi_\theta]$), which is an exploration technique used in policy gradient methods. Furthermore, the critic network is updated to minimize the state value (L^{VF}) difference. This process is expressed by Eqs. (2) to (4).

$$\text{maximize } L^{CLIP}(\theta) = \hat{E}_t[\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t)] \quad (2)$$

$$\text{minimize } L^{VF}(\theta) = R_a^s + \gamma V(s_{t+1}) - V(s_t) \quad (3)$$

$$\text{maximize } S[\pi_\theta] = -\sum \pi(a_t|s_t) \log \pi(a_t|s_t) \quad (4)$$

A technique for individual optimization is already provided in the advantage actor critic and PPO integrates this technique into the following equation for optimization:

$$\text{maximize } L^{CLIP+VF+S} = \hat{E}_t[L^{CLIP}(\theta) - c_1 L^{VF}(\theta) + c_2 S[\pi_\theta(s_t)]] \quad (5)$$

where c_1 and c_2 are hyperparameters that define the extent of value-

state network optimization and exploration. PPO outperforms TRPO and currently is the state-of-the-art RL algorithm using a policy gradient.

2.3. Related works

In the field of stock trading and asset allocation, RL algorithms are utilized in the financial market environment to maximize returns while minimizing risk. These algorithms learn from experience by exploring different actions and evaluating their outcomes as well as updating their policies accordingly. This approach enables RL algorithms to adapt to changing market conditions and generate more accurate predictions and investment decisions.

Previous research on RL-based asset allocation can be broadly categorized into two approaches: internal function conversion and fusion of other techniques. Internal function conversion primarily focuses on methods for enhancing the performance of a model in terms of returns, MDD, Sharpe ratio, and other metrics by modifying reward, loss, and objective functions. For example, Lee et al. (2020) proposed a multi-agent RL-based portfolio management system known as MAPS that combines correlation with a loss function to diversify portfolio configurations. This approach was inspired by the portfolio diversification strategies used by investment companies. Similarly, Li et al. (2007) proposed independent factor RL for portfolio management, which combines the weights of each factor with an optimization function using mean-variance optimization. Jiang et al. (2017) proposed an ensemble of identical independent evaluators that evaluates the future value of a portfolio and modifies portfolio weights accordingly. Hu et al. (2019) proposed finding an optimal policy by utilizing gated recurrent units in combination with a policy gradient.

Research on the fusion of other techniques focuses on methods for improving model performance by extracting variables affecting inputs and outputs or by combining various algorithms. For example, Soleymani and Paquet (2020) proposed a framework called DeepBreath that uses an autoencoder to expand input data and executes the SARSA algorithm for training either online or offline. Ye et al. (2020) proposed state-augmented RL, which integrates trend prediction results with various forms of data using an autoencoder. Finally, Betancourt and Chen (2021) proposed a method for integrating an algorithm that optimizes transaction costs with existing systems.

Most previous studies have focused on composing a portfolio based on novel asset allocation strategies to increase returns. In contrast, in this study, we aimed to minimize market risk by combining previously

developed PDAS instead of developing a new asset allocation strategy. Additionally, we propose a novel framework for overcoming the limitations of existing methods.

3. Proposed framework: Safety AARL

In this paper, we propose a novel framework called safety AARL that combines PDAS through RL to minimize risk. The proposed framework includes four major modules, as outlined in Fig. 2. The first module for data preprocessing prepares data for executing PDAS and data constituting an RL stock market environment. The submodules of the data preprocessing include portfolio data per PDAS, technical indicator generation for establishing PDAS, and data normalization. The second module for feature extraction compresses data and extracts the major variables to be used in RL training and testing. The third module for the RL environment interacts with the RL agent and delivers state and reward information according to received actions. The submodules of the RL environment include the Top3 softmax function, combine ratio reward function, PDAS function, train mode, and test mode. The fourth module for the RL agent interacts with the RL environment, trains the agent based on received state and reward information, and updates the agent's policy.

3.1. Data preprocessing

3.1.1. Portfolio data of each PDAS

A total of eight PDAS are used in the safety AARL framework. The assets belonging to a portfolio in each strategy vary, but some assets are common to all strategies. Fig. 3 presents the number of assets constituting a portfolio in each strategy. The minimum number of assets required in each strategy varies from 4 to 17 and the total number of unique assets is 24, which is the minimum number of assets required by the proposed framework.

Table 1 presents the results of matching the 24 unique assets with the symbols used on the Yahoo Finance website. Each asset consists of the US ETF used in each strategy and includes stocks, bonds, raw materials, real estate, and gold. The asset data used in this study include daily

values of opening, high, low, and closing prices, trading volume, and day of the week. Even though PDAS require monthly data, daily data were used to learn more diverse perspectives and then converted into monthly data.

3.1.2. Generating technical indicators to establish PDAS

Essentially, dynamic asset allocation strategies proceed with transactions based on sell and purchase rules. In this study, all strategies use the monthly rebalancing sell rules in which investment weights are adjusted every month; however, the purchase rules evaluate and compare the assets based on the technical indicators used in each strategy to decide the purchase of assets.

Table 2 presents a list of all intermediate processes and variables used for computing the technical indicators associated with each strategy. These technical indicators are computed identically within all assets to be used as future input variables. The moving average convergence divergence (MACD) is an indicator that represents the extent of simple and long-term moving averages converging or diverging. The signal lines for the 12 day simple and 26 day exponential moving averages use the 9 day exponential moving average of the MACD. The Bollinger band is an indicator that represents where a stock price is positioned with respect to the 20 day moving average and an upper band and lower band are used. The relative strength index (RSI), which represents the strength of the current trend as a percentage, is a useful indicator for predicting when a stock price trend will change. The RSI indicates relative strength based on the sum of rises and falls over 14 days. The commodity channel index (CCI) is a variability indicator that represents how far the stock price deviates from the average value over a certain period by calculating the deviation between the stock price and moving average. In this study, the 14 day moving average was used. The Dow Jones 30 industrial average (DOW 30) is a representative index of the US stock market based on the price fluctuations of 30 companies representing each sector among the companies listed on the New York Stock Exchange and NASDAQ. Although this is not a technical indicator, it is used to reflect the market trends of the US simple moving average (SMA) and is the most commonly used indicator for identifying the trends of stock prices. The closing price of a specific period is averaged,

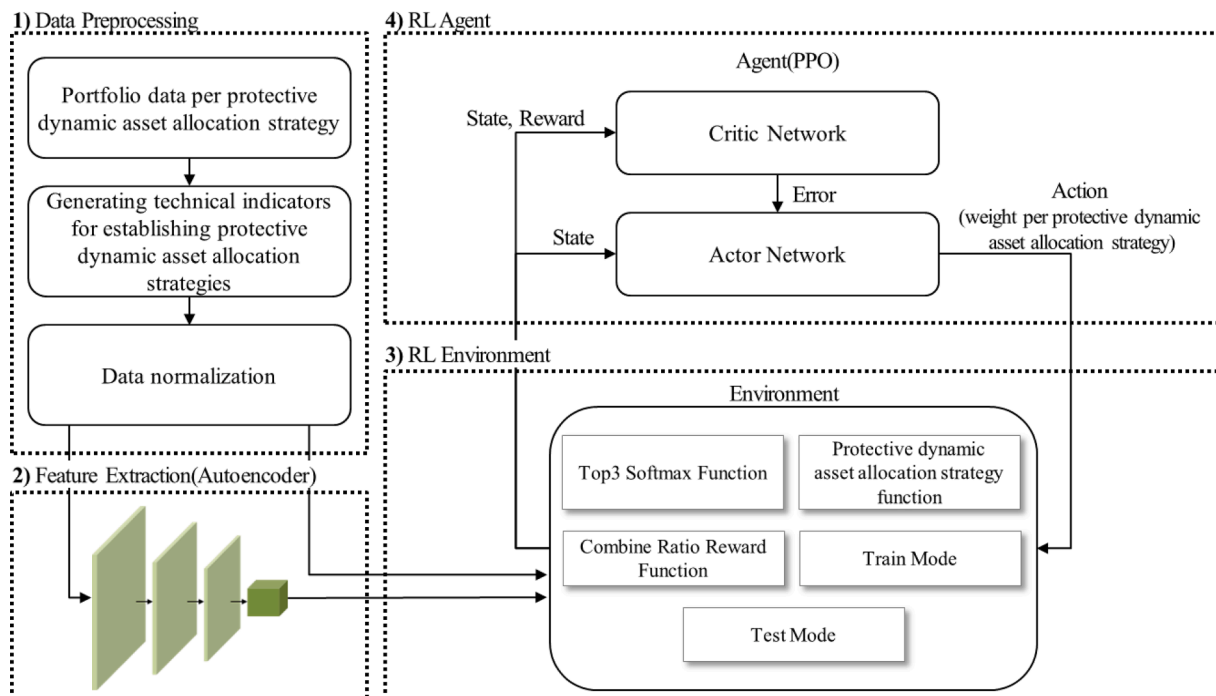


Fig. 2. Safety AARL architecture. The proposed framework includes four major modules: data pre-processing, feature extraction, an RL environment, and an RL agent.

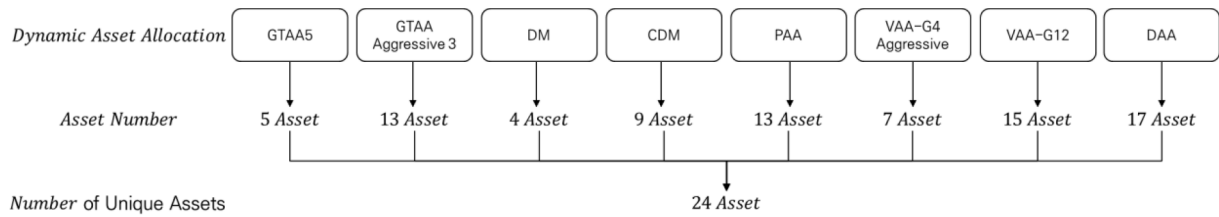


Fig. 3. Number of assets per PDAS.

Table 1

Symbols of the 24 unique assets considered in this study.

Asset Symbol				
AGG	BIL	BND	DBC	EEM
EFA	EWJ	GLD	HYG	IEF
IWD	IWM	LQD	QQQ	REM
SHY	SPY	TLT	VGK	VNQ
WO	^DJI	^FVX	^TYX	–

Table 2

Thirty-five variables processed by PDAS.

Basic Variables				
Open	High	Low	Close	Volume
Day of the week	–	–	–	–
Technical Indicators				
MACD	Boll UB	Boll LB	RSI	CCI
DOW 30	SMA 30D	SMA 60D	SMA 10 M	SMA 12 M
Close 1 M	Close 2 M	Close 3 M	Close 4 M	Close 5 M
Close 6 M	Close 7 M	Close 8 M	Close 9 M	Close 10 M
Close 11 M	Close 12 M	Return 1 M	Return 3 M	Return 6 M
Return 12 M	AVG Return	MOM 12 M	MOM Score	–

so it represents the overall price during that period. We considered 30 day, 60 day, 10 month, and 12 month SMAs. Close N-month simply represents the closing price of the N prior months and 12 periods ranging from 1 month to 12 months were considered. Return N-month represents the return calculated based on the current closing price and the closing price of the N prior months. It is an indicator that identifies the rise of the current closing price compared to the N prior months. A total of four periods (1, 3, 6, and 12 months) were considered. AVG Return is the average value of 1, 3, 6, and 12 month return values. As it represents the average returns over a period as long as the past year, it is an indicator that identifies overall short-, mid-, and long-term returns. Momentum N-month (MOM) is an indicator that identifies trends based on returns. Long-term trends were examined using a 12 month period. The MOM score represents trends over the past year by assigning different weights to short-, mid-, and long-term returns. It effectively reflects recent trends. Weights of 12, 4, 2, and 1 are multiplied by the 1, 3, 6, and 12 month returns, respectively, and then the average value is calculated. In this study, 35 variables were utilized, which consisted of 29 technical indicators and six basic variables, as defined in Table 2. These 35 variables retain their original values as non-normalized data and are transmitted to both the RL environment and data normalization module. They play a critical role in computing numerical values during the execution of each strategy in the RL environment and serve as the primary inputs for the data normalization module.

3.1.3. Data normalization

The stock data used in this study are time series data that must be normalized to the same scale because each asset has different prices. Performing normalization enables multidimensional values to be compared and analyzed easily, prevents overfitting and underfitting, and improves stability and convergence speed during optimization. The minmax scaler defined in Eq. (6) was used for normalization. The

minmax scaler readjusts data so that all values are between zero and one. All testing data were processed using a scaler defined from the training data.

$$x_{new} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (6)$$

Here, x is the variable value to be converted, x_{min} is the minimum value of the variable, x_{max} is the maximum value of the variable, and x_{new} is the converted x value. Normalization was executed to readjust a total of 35 variable values to a range of zero to one. The adjusted values were then inputted into the feature extraction module.

3.2. Feature extraction

The feature extraction module uses an autoencoder to optimize the training process and reduce data dimensions to extract variables without correlation (Liou et al., 2014). An autoencoder is a neural network that adds noise to inputs and attempts to reconstruct outputs. It consists of an encoder and decoder. The encoder expresses an input in reduced dimensions because the neurons in a hidden layer have smaller dimensions than inputs. The decoder is trained to reconstruct and output the features compressed to reduced dimensions in hidden layers. This process reduces redundancy and correlation while extracting extremely beneficial features (Lecun et al., 2015).

The architecture of the autoencoder used in this study is illustrated in Fig. 4. The network consists of a total of seven layers: one input layer, five hidden layers, and one output layer. The number of neurons in the input and output layers is calculated by multiplying the number of assets by the number of variables. Because 24 assets and 35 variables were used in this study, the total number of neurons is 840. The number of neurons in the first and fifth hidden layers is 500, and that in the second and fourth hidden layers is 300. The third neuron layer compresses the variables from the encoder into a bottleneck and reconfigures the inputs for the decoder based on the compressed variables. The number of neurons is calculated by multiplying the number of assets by five. The total number of neurons is 120 because 24 assets were used in this study. Once the autoencoder is trained, the decoder is discarded and only the encoder is retained so that only the variables compressed in the bottleneck are used for RL training. The autoencoder's parameter configurations included a learning rate of 0.01, leaky rectified linear unit as an activation function, mean squared error as a loss function, Adam as an optimization method, batch size of 16, and total of 3000 training epochs.

3.3. RL environment

The RL environment was constructed from data that were not normalized by the data preprocessing module and data consisting of features extracted by the feature extraction module. The RL environment interacts with the agent based on two types of received data and provides state and reward information according to received actions. The functions constituting the environment include the Top3 softmax function, dynamic asset allocation strategy function, combine ratio reward function, training mode, and testing mode. The Top3 softmax function readjusts the probabilities of the three actions with the largest

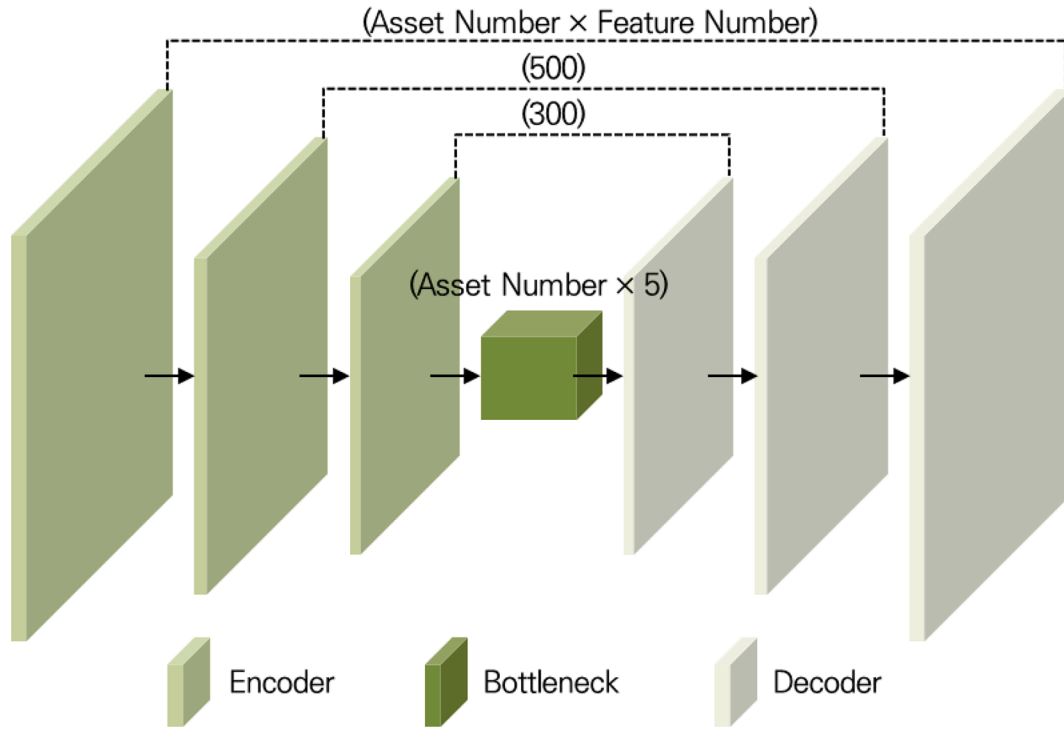


Fig. 4. Architecture of the autoencoder in the feature extraction module.

probabilities among the eight actions consisting of investment weights for DAS received from the agent. Eight matrices of zeros are created and actions are sorted in an ascending order to select the top three actions. Then, the sum of the probabilities of the next three actions is divided by three and inputted into the zero matrices. The sum of all probabilities is one, so only the top three actions have values other than zero. Next, the PDAS function, which plays the most important role in the proposed method, receives the actions readjusted by the Top3 softmax function and matches them with PDAS. Each strategy is matched with one action and the reward is calculated based on the investment weight or action value. The strategies that are not matched with the top three actions received from the Top3 softmax function receive no investment. Similar to the actions, each strategy is multiplied by the action weights to adjust the strategy weights because the overall investment weight is one. The buy rules for the eight PDAS were configured as shown in Table 3. The selling rules of the strategies are identical for monthly rebalancing.

When executing these strategies, the returns of the assets belonging to each strategy are individually calculated as shown in Equation (7) and then combined. Here, $Close_{sell}$ is the closing price at the time of selling and $Close_{buy}$ is the closing price at the time of buying. The return in a case where cash is held after executing a strategy is calculated to be zero.

$$AssetReturn = \left(\frac{Close_{sell}}{Close_{buy}} - 1 \right) \times Weight, \quad (7)$$

Next, the combine ratio reward function calculates the reward by summing the three performance indicators based on the returns calculated by the protective dynamic asset allocation strategy function. The three performance indicators are the Sharpe, Sortino, and Calmar ratios. The Sharpe ratio (Dowd, 2000) is a general ratio comparing reward (investment returns) and risk (standard deviations). It defines how well investment returns compensate for the risk borne by investors. As shown in Eq. (11), after subtracting the risk-free interest rate from the annual expected returns, the annual returns are divided by the standard deviation. The Sortino ratio (Sortino & Price, 1994) is very similar to the Sharpe ratio, but the Sharpe ratio calculates the standard deviation

using all returns, whereas the Sortino ratio only considers the standard deviation of declining returns. This ratio indicates that rising returns are favorable and declining returns are unfavorable. As shown in Eq. (12), after subtracting the risk-free interest rate from the expected annual returns, the annual returns are divided by the standard deviation. The Calmar ratio (Pinelis & Ruppert, 2022) compares returns with respect to the MDD instead of using standard deviations. As shown in Eq. (13), the expected annual returns are divided by the MDD. These three performance indicators are summed to determine the combine ratio, as shown in Eq. (14). The calculated combine ratio is given to the agent as a reward value to proceed with training.

$$\bar{R} = Expectedannualreturn \quad (8)$$

$$R_f = Annualrisk - freeinterestrate$$

$$\sigma = Annualreturnstandarddeviation \quad (9)$$

$$\sigma^- = Annualdeclinedreturnstandarddeviation \quad (10)$$

$$SharpeRatio = \frac{\bar{R} - R_f}{\sigma} \quad (11)$$

$$SortinoRatio = \frac{\bar{R} - R_f}{\sigma^-} \quad (12)$$

$$CalmarRatio = \frac{\bar{R}}{MaxDrawdown} \quad (13)$$

$$CombineRatio = SharpeRatio + SortinoRatio + CalmarRatio \quad (14)$$

The RL environment is defined differently in the training and testing modes. The training mode is used to train an RL agent. Training is performed by converting daily data into monthly data. An episode is divided into main episodes and training episodes to perform training using a large amount of data. Daily data are augmented with data from the previous 22 days to convert them into monthly data. A training

Table 3

Buy rules for the eight PDAS.

Strategy Name	Strategy Rule									
GTAA5 (Faber, 2007)	●	Buy Rule								
		✓	Invest 20% of assets into each ETF							
		✓	Hold each ETF if the price of each ETF is higher than the 10-month moving average, otherwise hold cash							
GTAA Aggressive 3 (Faber, 2007)	●	Buy Rule								
		✓	Calculate 1, 3, 6, and 12 month average returns of each ETF							
		✓	Invest the same amount of money in the ETFs with the three highest returns							
		✓	Hold cash if the price of each ETF is lower than the 10 month moving average							
DM (HA & Fabozzi, 2022)	●	Buy Rule								
		✓	Calculate the last 12 month return of the US stock SPY, developed market EFA, and ultra-short bonds BIL at the end of each month							
		✓	Invest in the ETF with the higher 12 month return between SPY or EFA if the return of SPY is higher than that of BIL							
		✓	Invest in AGG if the return of SPY is lower than that of BIL							
CDM (HA & Fabozzi, 2022)	●	Buy Rule								
		✓	Calculate the last 12 month return of two assets in each sector at the end of each month							
		✓	Invest in the ETF with the higher return							
		✓	Invest in the US dollar if the return of both ETFs is lower than that of BIL							
PAA (Keller & Keuning, 2016)	●	Buy Rule								
		✓	Set the weight of risk-free assets (breadth momentum)							
		➤	Find the number of assets with a lower current price than the 12 month SMA (declining) among 12 ETFs							
		➤	Risk-free asset weights are set as follows according to the number of declining ETFs:							
			No. of declining ETFs	0	1	2	3	4	5	6 or more
			Risk-free asset weight	0%	16.67%	33.33%	50%	66.67%	83.33%	100%
		✓	Invest in the US mid-term government bond (IEF) according to the weight of risk-free assets							
		✓	For the amount not invested in risk-free assets, diversely investments in six ETFs based on relative momentum							
		➤	Calculate the 12 month SMA of each ETF at the end of every month and invest in six assets with the highest value of (current price/12 month SMA)–1							
VAA-G4 Aggressive (Keller & Keuning, 2017)	●	Buy Rule								
		✓	Calculate the momentum scores of aggressive and risk-free assets at the end of every month							
		✓	Calculate the momentum score of each asset							
		➤	(12×1 month return)+(4×3 month return)+(2× 6month return)+(1×12 month return)							
		✓	If the momentum scores of all four aggressive assets are greater than zero, invest the entire portfolio in the aggressive asset with the highest momentum score							
		✓	If the momentum score of any of the four aggressive assets is below zero, invest the entire portfolio in the risk-free asset with the highest momentum score							
VAA-G12 (Keller & Keuning, 2017)	●	Buy Rule								
		✓	Set the weight of risk-free assets (breadth momentum)							
		➤	Find the number of (declining) assets with a current momentum score below zero among 12 ETFs of aggressive assets							
		➤	Momentum score: (12×1 month return)+(4×3 month return)+(2× 6 month return)+(1×12 month return)							
			No. of declining ETFs	0	1	2	3	4 or more		
			Risk-free asset weight	0%	25%	50%	75%	100%		
		✓	Invest in the risk-free asset with the highest momentum score according to the risk-free asset weight							
		✓	For the amount not invested in risk-free assets, invest in five ETFs by applying relative momentum							
		➤	Calculate the momentum score of each ETF at the end of every month and invest the same amount of money in the ETFs with the five highest scores							
DAA (Keller & Keuning, 2018)	●	Buy Rule								
		✓	Determine the weights of aggressive assets based on canary universe (VWO, BND)							
		➤	Momentum score of both VWO and BND is above zero: 100%							
		➤	Momentum score of either VWO or BND is above zero: 50%							
		➤	Momentum score of both VWO and BND is below zero: 0%							
		➤	Momentum score: (12×1 month return)+(4×3 month return)+(2× 6 month return)+(1×12 month return)							
		✓	Invest in the risk-free asset with the highest momentum score according to the risk-free asset weight							
	✓	For the amount not invested in risk-free assets, invest in six ETFs by applying relative momentum								
	✓	Calculate the momentum score of each ETF at the end of every month and invest the same amount of money in the ETFs with the six highest scores								

episode consists of 22 epochs in which monthly data are created by augmenting with data one day at a time from the starting date. After 23 days, when the data begin to overlap, the main episode is augmented with one day. One main episode consists of 22 training episodes and once an entire main training episode is completed, the main episode is augmented by an additional day for further training. Accordingly, the

numbers of daily data and monthly data become identical. The testing mode is used to test the trained RL agent. Unlike the training mode, the testing mode does not involve training episode. Daily data are received and augmented with 22 days from the starting date to convert them into monthly data. The weights of the PDAS are recorded for the agent as it receives each state and the returns generated by those weights are

recorded.

3.4. RL agent

In this study, an RL agent was used to adjust the investment weights of PDAS. The RL agent interacts with an environment and outputs actions based on received states and rewards. The RL agent is trained based on the PPO algorithm and the architecture of the policy network is illustrated in Fig. 5. The input embedder flattens a two-dimensional state (number of assets: 5) into one dimension. The shared network delivers the state that has been flattened to the actor and critic networks. The actor and critic networks (Konda & Tsitsiklis, 1999) have the same architecture, except for their output layers. The actor network selects an action $\pi(s,a)$ when a state is given, whereas the critic network evaluates the value of a state $V(s)$. The RL agent uses the data extracted by the feature extraction module from the RL environment as inputs and it outputs the weight of eight PDAS. Each weight has a real number value between zero and one.

4. Experiments

Three experiments were conducted to verify the effectiveness of the proposed safety AARL framework. In the first experiment, the performance of the framework was used to compare the eight PDAS in the case of investing with equal weights in all eight strategies during a test period

from July 1 of 2018 to March 31 of 2022 for bull, flat, and bear markets. The second experiment was conducted to verify the performance of safety AARL during three periods when the global stock market performed particularly poorly. The third experiment compared the performance of the proposed framework to that of the latest asset allocation methods using RL.

4.1. Performance evaluation method

Annual and cumulative returns, the Sharpe ratio, and MDD were used as performance evaluation indicators. Annual returns are the weighted averages of returns on assets during corresponding periods. Cumulative returns are calculated by subtracting the final value from the initial value and then dividing the result by the initial value. The Sharpe ratio is calculated by subtracting the annual risk-free interest rate from annual returns and then dividing the result by the annual variability. The MDD represents the maximum loss ratio during the transaction period, where a lower value indicates better performance.

4.2. Experimental environment

The experimental environment used in this study is outlined in Table 4. For the feature extraction module, Keras (Ketkar, 2017) was used as autoencoder. The RL environment was constructed using Open AI Gym (Vidyardhar, Nagaraj, & Ashoka, 2021). PyTorch (Paszke et al.,

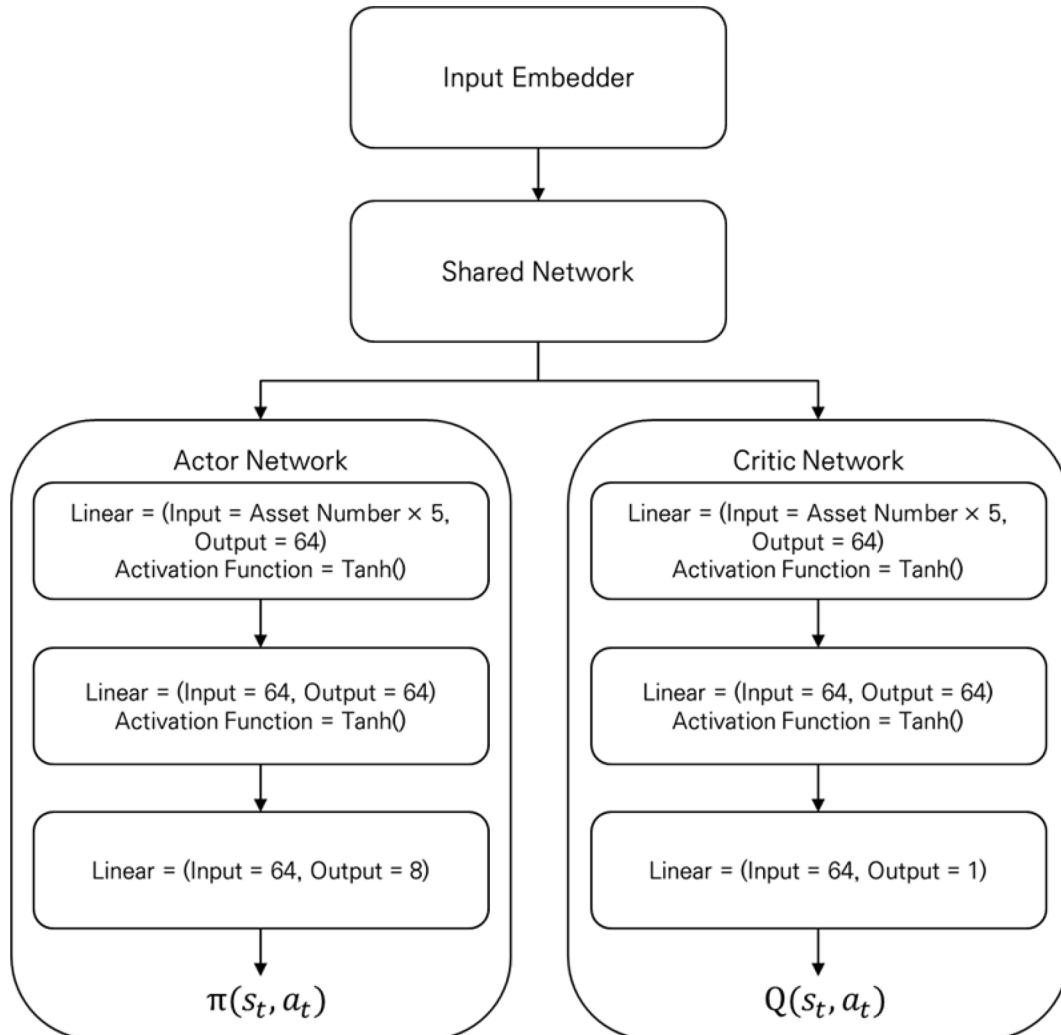


Fig. 5. Architecture of PPO actor-critic network.

Table 4

Experimental Environment for all experiments.

Device	Specifications
OS	Windows 10
CPU	Intel Core i9-9900KF 3.6 GHz
GPU	NVIDIA GeForce RTX 2080Ti × 2
RAM	128 GB
Storage	2 TB SSD
Language	Python 3.6.13, PyTorch = 1.8.1, Stable-Baselines3 = 1.3.0, Tensorflow = 2.1.0, Keras = 2.3.1, Open AI gym = 0.19.0

2019) and Stable-Baselines3 (Raffin et al., 2021) were used as RL agents. In our experiments, the model training duration and number of epochs were defined as follows. The autoencoder required 1.62 s per epoch and the total time elapsed for 3000 epochs was 1 h and 21 min. The RL component required 2.45 s per epoch and the training period for 1310 epochs lasted approximately 54 min. Although RL attempted to perform 2000 epochs in total, the early stop function terminated the training prematurely at epoch 1310.

4.3. Performance comparison during test periods

To verify the effectiveness of safety AARL, Experiment 1 compared the performance of eight PDAS for which stability and performance have previously been proven by investing equally into the eight strategies. The assets used for this experiment are listed in Table 1. Data from January 1 of 2009 to June 30 of 2018 were used as training data and data from July 1 of 2018 to March 31 of 2022, were used as testing data. The eight PDAS compared in this study were established through trial and error by experts who specialize in dynamic asset allocation. Furthermore, the validity of adjusting weights for each strategy was verified by comparing the results to those of the method of investing with equal weights in all eight strategies in the same environment as safety AARL. The experimental results are presented in Table 5. Safety AARL outperformed all strategies excluding GTAA Aggressive3 (Faber, 2007) on the two performance indicators related to returns and outperformed all strategies on the Sharpe ratio and MDD performance

Table 5

Performance comparisons during testing periods for Experiment 1.

System	Annual Return	Cumulative Returns	Sharpe Ratio	Max Drawdown
Dow Jones Index	9.097	38.613	0.580	23.201
Equal Weight	7.251	30.017	1.071	4.533
GTAA5 (Faber, 2007)	7.352	30.480	1.011	5.248
GTAA Aggressive 3 (Faber, 2007)	11.456	50.187	0.921	16.631
DM (HA & Fabozzi, 2022)	5.997	24.408	0.442	19.448
CDM (HA & Fabozzi, 2022)	0.559	2.114	0.121	7.499
PAA (Keller & Keuning, 2016)	8.088	33.867	1.226	6.890
VAA-G4 Aggressive (Keller & Keuning, 2017)	7.127	29.455	0.775	10.077
VAA-G12 (Keller & Keuning, 2017)	7.787	32.474	1.061	11.075
DAA (Keller & Keuning, 2018)	7.898	32.985	1.087	8.053
Safety AARL (Ours)	10.259	44.229	1.350	3.828

indicators, which represent stability. The Sharpe ratio was 10.114% higher than the highest PAA (Keller & Keuning, 2016) strategy. The MDD was 18.417% lower than the lowest equal weight. The proposed safety AARL framework maximizes the effects of diversified investment and the characteristics of dynamic asset allocation strategies, which provides stability and the possibility of improving performance by combining the advantages of multiple strategies.

Fig. 6 presents the cumulative returns graph for each strategy during the testing period. Considering the market conditions with respect to the Dow Jones index in the graph, a flat market is initially maintained (July of 2018 to January of 2020), followed by a bear market with a 23.201% decline (February of 2020 to May of 2020), and then a bull market (May of 2020 to March of 2022). In the cumulative returns graph, safety AARL yields the smallest range of fluctuation, prevents drawdown in the bear market, and steadily rises in the bull market. This demonstrates that the proposed framework outperforms other PDAS that focus on stability and minimizes risk under various market conditions. Additionally, the proposed framework also provides superior performance in all areas compared to the equal weight strategy, which demonstrates its validity for adjusting investment weights for different strategies.

4.4. Performance verification in the falling global markets

In Experiment 2, performance verification was conducted in the well-known falling global markets to verify that the proposed framework can minimize market risks without being limited to the test data. The three falling global markets that were tested were the global financial crisis, China-US trade war, and COVID-19 pandemic periods. To conduct our performance assessment, six representative indexes were selected as benchmarks, namely the Dow Jones Index, Nikkei 225 Index, SSE Index, FTSE 100 Index, DAX Index, and CAC 40 Index.

4.4.1. Performance verification during the global financial crisis

The global financial crisis refers to the period in which a large-scale financial crisis started in the US finance market in the late 2000s and eventually spread worldwide. This was a period of global economic crisis with a similar level of impact as the Great Depression of 1929. Table 6 presents the results of this verification. Data from February 1 of 2010 to March 31 of 2022 were used as the training data and data from January 1 of 2008 to January 1 of 2010 were used as testing data. Excluding the SSE index, all other indices exhibit cumulative returns losses over the past 18 months, indicating unfavorable market conditions overall. Furthermore, the market exhibited sharp price fluctuations and poor performance, as evidenced by the negative Sharpe ratio value and average MDD value of 38.806%. In contrast, the proposed framework yielded over 18% annual returns, even during the bear market, while the Sharpe ratio and MDD were 1.349 and 8.301%, respectively, representing minimal market risk.

Fig. 7 shows the cumulative return graph during the period of global financial crisis. The graph shows that the market continued to fall from the end of 2018 and recorded its lowest point in March 2019. The graph corresponding to Safety AARL shows that a loss was incurred in March 2019, but the risk was less than that of the market. In addition, all returns from September 2008 to January 2009, when the market abruptly declined, were realized, and the overall cumulative return steadily increased.

4.4.2. Performance verification during the China-US trade war

The China-US trade war began at the end of 2018 and an agreement was reached in January of 2020 after prolonged economic conflict between the two countries. As a result of economic stress during that period, the market continuously declined based on weak investor sentiment. Table 7 presents the performance of the proposed framework based on data from the China-US trade war period. The data from January 1 of 2009 to June 30 of 2018 were used as training data and the data from August 1 of 2018 to August 1 of 2019 were used as testing

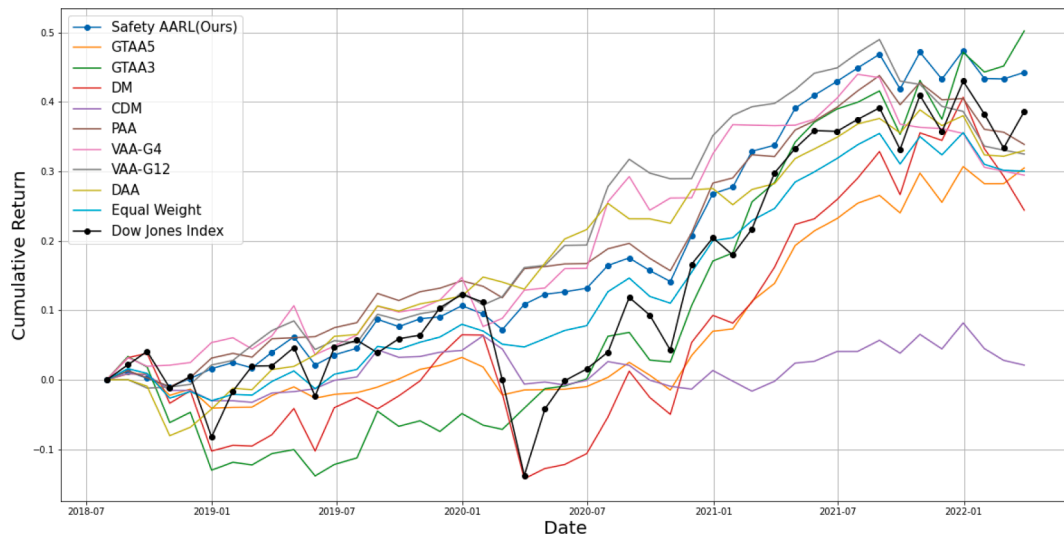


Fig. 6. Cumulative return graph per strategy during the test period for Experiment 1.

Table 6

Performance verification during the global financial crisis for Experiment 2.

System	Annual Return	Cumulative Returns	Sharpe Ratio	Max Drawdown
Dow Jones Index	-5.210	-8.123	-0.131	38.817
Nikkei 225 Index	-14.364%	-21.770%	-0.362	43.860%
SSE Index	12.071%	19.774%	0.496	37.717%
FTSE 100 Index	-2.408%	-3.786%	-0.008	32.049%
DAX Index	-4.597%	-7.181%	-0.042	40.679%
CAC 40 Index	-7.255%	-11.241%	-0.191	39.712%
Safety AARL (Ours)	18.742	31.257	1.349	8.301

data. The proposed framework generated lower returns compared to both the Dow Jones and SSE indices. Regardless, the proposed framework yielded a Sharpe ratio of 0.56 and MDD of 3.828%, demonstrating its potential to minimize market risk.

Fig. 8 presents the cumulative returns graph during the China-US trade war period. The graph reveals that the market continuously

declined from the end of 2018 to January of 2019, where it recorded its lowest point, and then returned to a bull market. In contrast, the graph of safety AARL reveals that a loss was not incurred in January of 2019 and that risk was prevented while realizing returns. Excluding one day in May of 2019, when the value fell, the overall cumulative returns steadily increased.

Table 7

Performance verification during the China-US trade war for Experiment 2.

System	Annual Return	Cumulative Returns	Sharpe Ratio	Max Drawdown
Dow Jones Index	3.464	3.464	0.277	11.833
Nikkei 225 Index	-5.876%	-5.876%	-0.222	17.020%
SSE Index	7.605%	7.605%	0.461	11.606%
FTSE 100 Index	2.077%	2.077%	0.244	10.414%
DAX Index	-1.416%	-1.416%	-0.014	14.600%
CAC 40 Index	2.072%	2.072%	0.200	13.886%
Safety AARL (Ours)	3.264	3.264	0.560	3.828

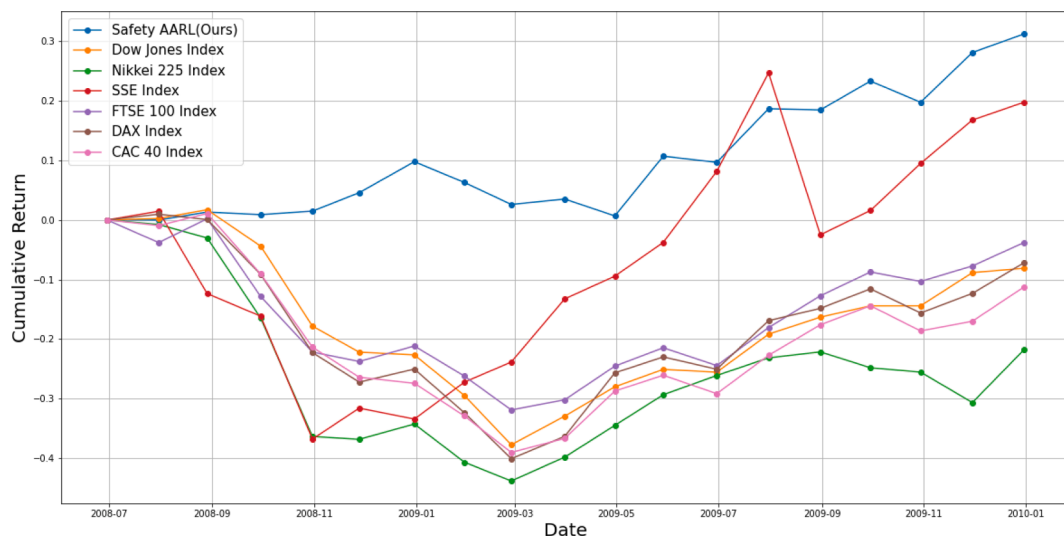


Fig. 7. Cumulative returns for each strategy during the global financial crisis for Experiment 2.

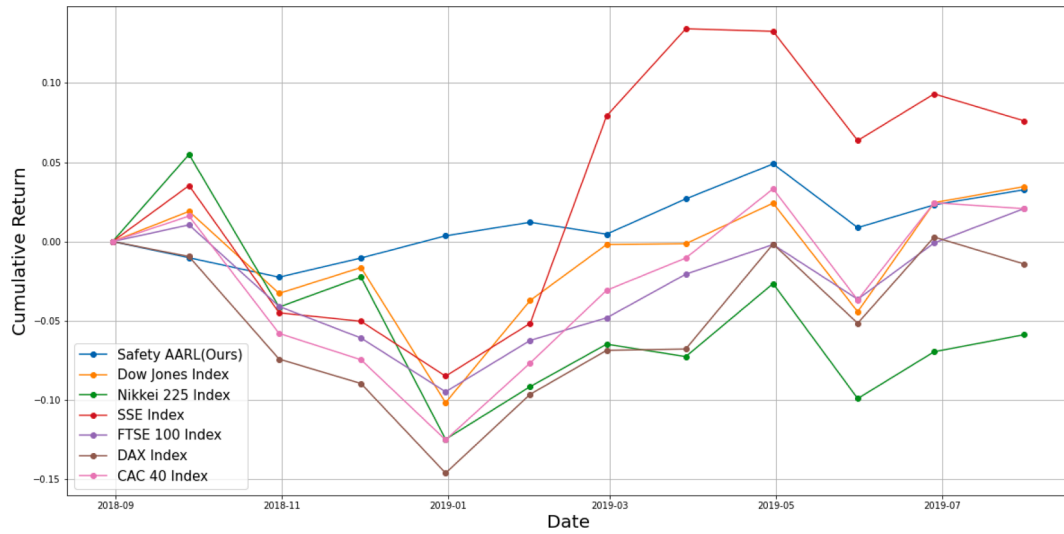


Fig. 8. Cumulative returns of the global indexes during the China-US trade war for Experiment 2.

4.4.3. Performance verification during the pandemic

The COVID-19 pandemic was a time of significant anxiety and uncertainty, and was particularly devastating from early March to early April of 2020, when the market abruptly declined as the World Health Organization declared the outbreak of the coronavirus to be a pandemic. The highest warning level was issued on March 11 of 2020. Table 8 presents the results of verifying the performance of the proposed framework during the COVID-19 pandemic. Data from January 1 of 2009 to June 30 of 2018 were used as training data and data from November 1 of 2019 to May 1 of 2020 were used as testing data. All indices, including the SSE index, which had performed well in the previous two bear markets, suffered losses over this six month period, indicating unfavorable overall market conditions. Additionally, the market exhibited significant price fluctuations and underperformance, as evidenced by negative Sharpe ratio values and high MDD values. In contrast, the proposed framework provided annual returns of over 6%, even during the bear market, with a Sharpe ratio and MDD of 0.895 and 3.102%, respectively, highlighting its potential to mitigate market risk.

Fig. 9 presents the cumulative returns graph for the period of the COVID-19 pandemic. One can see that the declining trend lasted from approximately February to April of 2020. In contrast, the graph of safety AARL reveals that loss was minimized and prevented during the declining period, and that the overall cumulative returns increased.

The performance of the proposed framework has been verified for three global markets with a noticeable decline; the return of the proposed framework steadily increased regardless of market conditions and the market risk was minimized. These results demonstrate that the proposed framework is not limited to the test data, and that the deviation in performance is not significant when using new test data or

according to market conditions.

4.5. Comparison of the latest research

In Experiment 3, the performances of the proposed framework and frameworks developed in studies with a high citation frequency (from the latest research on asset allocation using RL) were compared. Table 9 presents the results when setting the testing periods of all frameworks to be identical to the testing period of the proposed framework. Different market environments and asset types were used in this experiment, but this was presumed to have no impact because only investment performance was measured. Wu et al. (2021) proposed a portfolio management system (PMS) that uses RL by configuring an agent with two neural networks (convolutional neural network (CNN) and recurrent neural network (RNN)) and reducing risk by using the Sharpe ratio as a reward. This comparison was conducted on data representing a period of approximately two years and PMS_CNN was compared to PMS_RNN. The experimental results revealed that PMS_CNN and PMS_RNN had 5.295 and 1.939 times higher returns, respectively, than safety AARL. However, safety AARL had a 2.583 times and 3.464 times higher Sharpe ratio, and 7.902 times and 3.998 times lower MDD than PMS_CNN and PMS_RNN, respectively. Soleymani and Paquet (2020) proposed a framework called DeepBreath that can extract input data through an autoencoder and perform online or offline separately training using the SARSA algorithm. This comparison was conducted on data from a period of approximately four months. The experimental results revealed that DeepBreath had 2.091 times higher returns than safety AARL. However, Safety AARL had a 1.471 times higher Sharpe ratio and 49.881 times lower MDD.

Overall, the experimental results revealed that the latest frameworks yield higher returns than the proposed framework, but exhibit a lower Sharpe ratio and higher MDD, indicating insufficient transaction stability. Safety AARL, which focuses on stability, yielded half the returns of the latest frameworks with a 2.506 times higher Sharpe ratio and 20.594 times lower MDD on average. This demonstrates that the proposed framework minimizes risk while guaranteeing an upward trend in returns.

5. Conclusion and future work

In this paper, we proposed a novel framework called safety AARL that minimizes risk by combining different PDAS based on RL. The proposed framework consists of four modules for data preprocessing, feature extraction, an RL environment, and RL agent. First, technical

Table 8

Performance verification during the COVID-19 pandemic for Experiment 2.

System	Annual Return	Cumulative Returns	Sharpe Ratio	Max Drawdown
Dow Jones Index	-24.676	-13.210	-0.781	23.201
Nikkei 225 Index	-24.847%	-13.309%	-1.151	20.035%
SSE Index	-0.827%	-0.414%	0.005	9.830%
FTSE 100 Index	-35.476%	-19.673%	-1.643	24.798%
DAX Index	-32.663%	-17.941%	-1.154	25.007%
CAC 40 Index	-40.051%	-22.573%	-1.732	26.462%
Safety AARL (Ours)	6.031	2.971	0.895	3.102

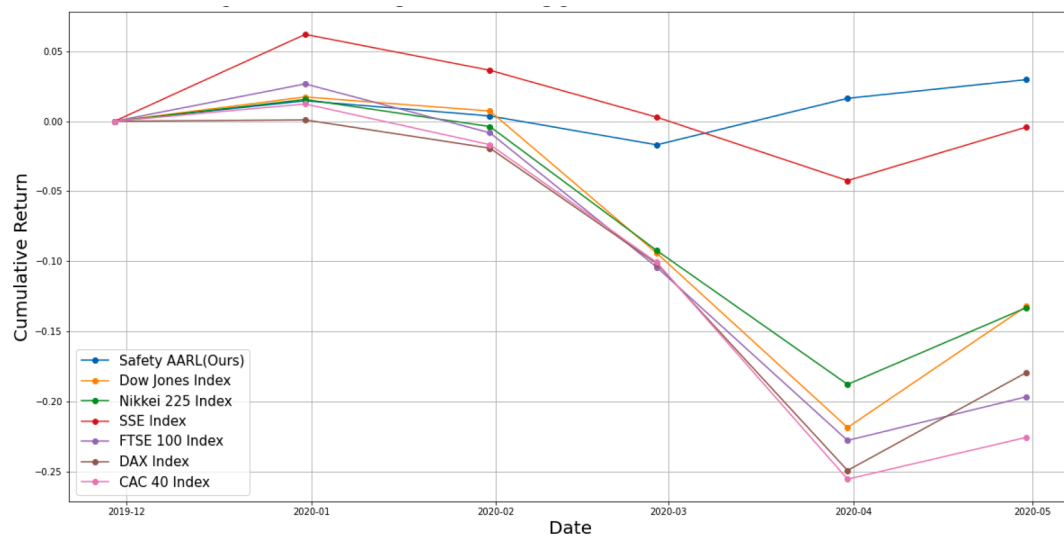


Fig. 9. Cumulative return of the global index during the COVID-19 pandemic for Experiment 2.

Table 9

Performance indicators for state-of-the-art frameworks and the proposed framework in Experiment 3.

System	Annual Return	Cumulative Returns	Sharpe Ratio	Max Drawdown
Test period: 2017-08-01 ~ 2019-07-31				
PMS_CNN (Wu et al., 2021)	48.888	103.041	0.578	17.843
PMS_RNN (Wu et al., 2021)	17.903	37.735	0.431	9.028
Safety AARL (Ours)	9.232	19.316	1.493	2.258
Test period: 2018-03-19 ~ 2018-07-27				
DeepBreath (Soleymani & Paquet, 2020)	29.585	11.930	2.260	12.520
Safety AARL (Ours)	14.152	5.669	3.325	0.251

indicators are generated and normalized based on the assets belonging to each strategy to execute PDAS in the data preprocessing module. The normalized data are then compressed and variables are identified by an autoencoder in the feature extraction module to address the problem of the number of input data increasing proportionally to the number of assets. The data generated by these two modules are inputted into the RL environment module. The RL environment module invests in the PDAS matched with three actions that received high action probabilities from the RL agent and interacts with the RL agent by delivering state and reward information, which are calculated based on a combination of the Sharpe, Sortino, and Calmar ratios. The RL agent receives the state and reward information from the RL environment and learns using the PPO algorithm, where the output actions are the investment weights for each PDAS. Actions are defined as PDAS to maintain their characteristics and the effects of diversified investments are determined by learning with various combinations of PDAS suitable for different market conditions. Three experiments were conducted to verify the effectiveness of the proposed framework. The experimental results demonstrated that the proposed framework minimizes risk, regardless of market conditions, and guarantees steady returns, indicating its excellent stability. Using the proposed safety AARL framework maximizes the effects of diversified investment and the characteristics of PDAS, thereby achieving stable asset allocation. However, the proposed Safety AARL framework has limitations in terms of the number of assets that can be used for PDAS and the performance of PDAS themselves. In future studies, an RL agent

will be developed to allow investment professionals to provide their perspective on the market based on initial trial and error, and make investments accordingly.

CRedit authorship contribution statement

Da Woon Jeong: Conceptualization, Methodology, Software, Validation, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Seong Joon Yoo:** Writing – review & editing. **Yeong Hyeon Gu:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government(MSIT) (No. 2022-0-00204, Development of AI based Autonomous Quality Control in Manufacturing).

References

- Almahdi, S., & Yang, S. Y. (2017). An adaptive portfolio trading system: A risk-return portfolio optimization using recurrent reinforcement learning with expected maximum drawdown. *Expert Systems with Applications*, 87, 267–279.
- Betancourt, C., & Chen, W.-H. (2021). Deep reinforcement learning for portfolio management of markets with a dynamic number of assets. *Expert Systems with Applications*, 164, 114002.
- Chen, Y.-Y., et al. (2021). Adversarial attacks against reinforcement learning-based portfolio management strategy. *IEEE Access*, 9, 50667–50685.
- Cheong, D., et al. (2017). Using genetic algorithm to support clustering-based portfolio optimization by investor information. *Applied Soft Computing*, 61, 593–602.
- Dowd, K. (2000). Adjusting for risk: An improved sharpe ratio. *International Review of Economics & Finance*, 9(3), 209–222.
- Faber, M. T. (2007). A quantitative approach to tactical asset allocation. *The Journal of Wealth Management*, 9(4), 69–79.
- Ha, S., & Fabozzi, F. J. (2022). Dual momentum: Testing the dual momentum strategy and implications for lifetime allocations. *The Journal of Portfolio Management*, 48(4), 282–301.

- Heaton, J. B., Polson, N. G., & Witte, J. H. (2017). Deep learning for finance: Deep portfolios. *Applied Stochastic Models in Business and Industry*, 33(1), 3–12.
- Hirata, T., et al. (2016). Deep belief network using reinforcement learning and its applications to time series forecasting. In *International Conference on Neural Information Processing* (pp. 30–37). Cham: Springer.
- Hu, Y.-J., & Lin, S.-J. (2019). Deep Reinforcement Learning for Optimizing Finance Portfolio Management. In *2019 Amity International Conference on Artificial Intelligence (AICAI)*.
- Jiang, Z., Xu, D., & Liang, J. (2017). A deep reinforcement learning framework for the financial portfolio management problem. arXiv preprint arXiv:1706.10059.
- Keller, W. J., & Keuning, J. W. (2016). Protective asset allocation (PAA): a simple momentum-based alternative for term deposits. Available at SSRN 2759734.
- Keller, W. J., & Keuning, J. W. (2017). Breadth Momentum and Vigilant Asset Allocation (VAA): Winning More by Losing Less. Available at SSRN 3002624.
- Keller, W. J., & Keuning, J. W. (2018). *Breadth Momentum and the Canary Universe: Defensive Asset Allocation (DAA)*. Available at SSRN 3212862.
- Ketkar, N. (2017). Introduction to Keras. *Deep Learning with Python*, 97–111.
- Konda, V., & Tsitsiklis, J. (1999). Actor-critic algorithms. *Advances in Neural Information Processing Systems*, 12.
- Lecun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436–444.
- Lee, R. S. T. (2019). COSMOS trader—Chaotic Neuro-oscillatory multiagent financial prediction and trading system. *The Journal of Finance and Data Science*, 5(2), 61–82.
- Lee, J., Kim, R., Yi, S.-W., & Kang, J. (2020). MAPS: Multi-Agent reinforcement learning-based Portfolio management System. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence*.
- Lei, K., et al. (2020). Time-driven feature-aware jointly deep reinforcement learning for financial signal representation and algorithmic trading. *Expert Systems with Applications*, 140, 112872.
- Li, J., Zhang, K., & Chan, L. (2007). Independent factor reinforcement learning for portfolio management. In *International Conference on Intelligent Data Engineering and Automated Learning* (pp. 1020–1031). Berlin, Heidelberg: Springer.
- Li, J., Zhang, K., & Chan, L. (2007). Independent Factor Reinforcement Learning for Portfolio Management. *Lecture Notes in Computer Science*, 1020–1031.
- Liou, C.-Y., et al. (2014). Autoencoder for words. *Neurocomputing*, 139, 84–96.
- Malkiel, B. G. (1989). Efficient market hypothesis. In *Finance* (pp. 127–134). London: Palgrave Macmillan.
- Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77–91.
- Park, H., Sim, M. K., & Choi, D. G. (2020). An intelligent financial portfolio trading strategy using deep Q-learning. *Expert Systems with Applications*, 158, 113573.
- Paszke, A., et al. (2019). Pytorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems*, 32.
- Pendharkar, P. C., & Cusatis, P. (2018). Trading financial indices with reinforcement learning agents. *Expert Systems with Applications*, 103, 1–13.
- Pinelis, M., & Ruppert, D. (2022). Machine learning portfolio allocation. *The Journal of Finance and Data Science*, 8, 35–54.
- Ponomarev, E. S., Oseledets, I. V., & Cichocki, A. S. (2019). Using reinforcement learning in the algorithmic trading problem. *Journal of Communications Technology and Electronics*, 64(12), 1450–1457.
- Prollochs, N., Feuerriegel, S., & Neumann, D. (2016). Detecting Negation Scopes for Financial News Sentiment Using Reinforcement Learning. In *2016 49th Hawaii International Conference on System Sciences (HICSS)*.
- Raffin, A., et al. (2021). Stable-baselines3: Reliable reinforcement learning implementations. *Journal of Machine Learning Research*, 22.
- Schulman, J., et al. (2017). Proximal policy optimization algorithms. arXiv preprint arXiv:1707.06347.
- Schulman, J., et al. (2015). Trust region policy optimization. In *32nd International Conference on Machine Learning, ICML 2015*, 3, pp. 1889–1897.
- Soleymani, F., & Paquet, E. (2020). Financial portfolio optimization with online deep reinforcement learning and restricted stacked autoencoder—DeepBreath. *Expert Systems with Applications*, 156, 113456.
- Sortino, F. A., & Price, L. N. (1994). Performance measurement in a downside risk framework. *The Journal of Investing*, 3(3), 59–64.
- Tversky, A., & Kahneman, D. (1992). Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and uncertainty*, 5(4), 297–323.
- Van Otterlo, M., & Wiering, M. (2012). Reinforcement learning and markov decision processes. In *Reinforcement learning* (pp. 3–42). Berlin, Heidelberg: Springer.
- Vidyadhar, V., Nagaraj, R., & Ashoka, D. V. (2021). NetAI-Gym: Customized environment for network to evaluate agent algorithm using reinforcement learning in open-AI gym platform. *International Journal of Advanced Computer Science and Applications*, 12 (4).
- Wu, M.-E., et al. (2021). Portfolio management system in equity market neutral using reinforcement learning. *Applied Intelligence*, 51(11), 8119–8131.
- Yang, S. Y., Yu, Y., & Almahdi, S. (2018). An investor sentiment reward-based trading system using Gaussian inverse reinforcement learning algorithm. *Expert Systems with Applications*, 114, 388–401.
- Yang, L. (2019). Loss aversion in financial markets. *Journal of Mechanism and Institution Design*, 4(1), 119–137.
- Ye, Y., et al. (2020). Reinforcement-learning based portfolio management with augmented asset movement prediction states. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pp. 1112–1119.
- Zhang, Y., et al. (2020). Cost-sensitive portfolio selection via deep reinforcement learning. *IEEE Transactions on Knowledge and Data Engineering*.